Spatial Estimates of SWE for the Rio Grande Basin

April 29, 2025

Max Gersh[1], Logan Stephenson[2], Ross Palomaki[2], and Karl Rittger[2]
[1] New Mexico Office of the State Engineer (NM OSE) Hydrology Bureau
[2] Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder
Work funded by: NASA Western Water Applications Office (WWAO)
Contact: Max.Gersh@ose.nm.gov

Overview Map – Rio Grande Basin Study Area

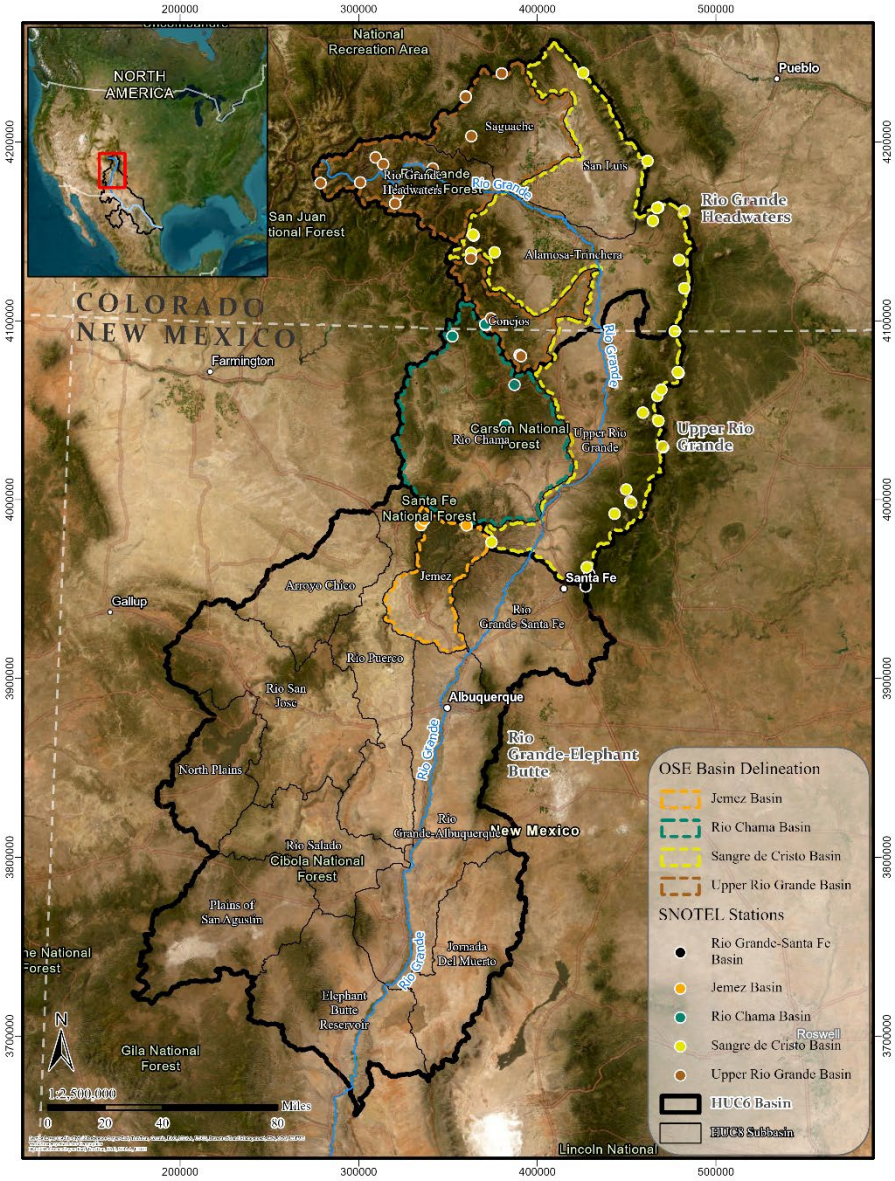

Figure 1. Overview map of the Rio Grande (RG) Basin study area. *The Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte Basins are presented as the standard HUC6 basins. HUC8 groupings (Jemez – orange, Rio Chama – green, Sangre de Cristo – yellow, and Upper Rio Grande – brown) are symbolized.*

Summary of SNODAS SWE for April 29, 2025

- Spatial estimates of snow water equivalent (SWE) and change in SWE (presented in inches) for the Rio Grande (RG) Basin study area are presented in Figure 2.
- Spring snowmelt persisted throughout the Rio Grande Basin study area, resulting in a decrease in SWE in all three HUC6 Basins since the last report (April 22, 2025).
    - Estimated SNODAS SWE decreased across all HUC8 subbasins and elevation bands except for the Upper Rio Grande HUC8 subbasin between 12,000 and 13,000 feet, which showed an increase of 338 acre-feet (Figure 4; Table 3).
    - Estimated SNODAS SWE decreased to 0 acre-feet throughout the entire Rio Grande-Elephant Butte HUC6 Basin (Figure 5; Tables 1 and 3).
- **Table 1**: Estimated SWE amount (median and mean in inches (in)), estimated volume (acre-feet (af)), and change in volume (af) since the last report, are shown for the standard Hydrologic Unit Code 6 (HUC6) basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Table 2**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown for the specific HUC8 subbasin groupings (OSE Basin Delineations shown in Figure 1; Upper Rio Grande, Rio Chama, Sangre de Cristo, and Jemez).
- **Table 3**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown by basin and elevation band for the HUC6 basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Rio Grande Headwaters (HUC6)**: SNODAS SWE volume for April 29, 2025, is estimated to be 247,158 af, a decrease of an estimated 247,857 af from April 22, 2025.
- **Upper Rio Grande (HUC6)**: SNODAS SWE volume for April 29, 2025, is estimated to be 62,532 af, a decrease of an estimated 57,952 af from April 22, 2025.
- **Rio Grande-Elephant Butte (HUC6)**: SNODAS SWE volume for April 29, 2025, is estimated to be 0 af, a decrease of an estimated 5,219 af from April 22, 2025.

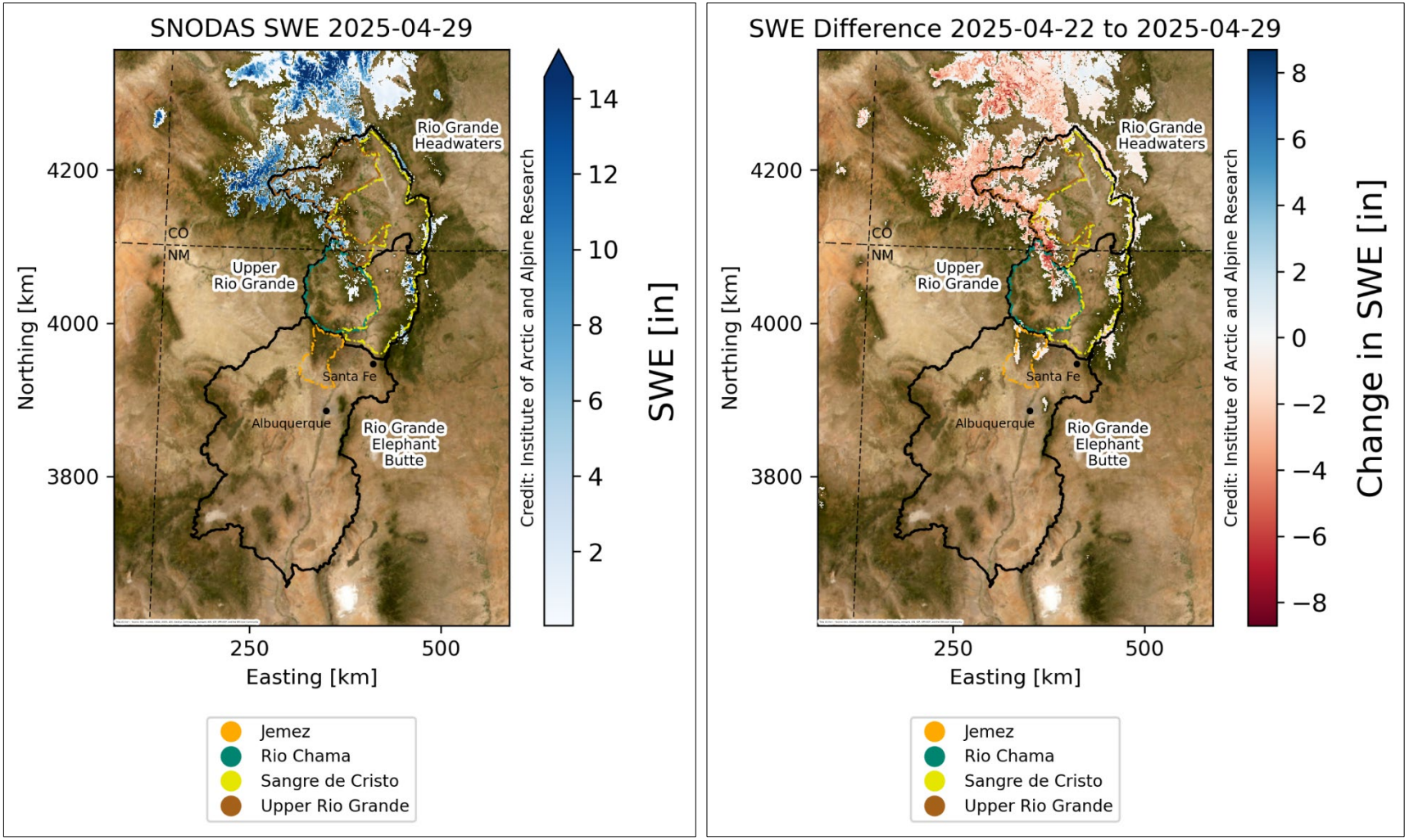

Figure 2. Estimated SNODAS SWE and Change in SNODAS SWE across the Rio Grande Basin study area. *SWE amounts (in) across the RG Basin study area (left); dark blue pixels denote greater SWE amounts. Change in SNODAS SWE amounts (in) across the RG Basin study area from April 22, 2025, to April 29, 2025 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Introduction

The Rio Grande (RG) is a snowmelt-driven river system with up to 75% of its water originating from seasonal snowpack in the mountains of Colorado and northern New Mexico (Moeser et al., 2021; Figure 2). Snowpack acts as a natural reservoir in the RG Basin, gradually releasing meltwater during the spring and early summer months, sustaining streamflow and providing vital water supply for downstream users. However, regional decreases in snowpack volume and earlier snowmelt timing, especially throughout the western U.S. and New Mexico, can lead to reduced water availability (Mote et al., 2005; EPA, 2016b; Mote et al., 2018; Hale et al., 2023). Long-term changes in snowpack patterns and timing can serve as an indicator of climate change, while understanding short-term changes is critical for monitoring seasonal variability and forecasting seasonal water supply. Monitoring snowpack provides valuable data on shifts in precipitation patterns and temperatures, helping scientists and decision-makers better understand and manage climate-related challenges.

Snow water equivalent (SWE) is the estimate of the depth of liquid water contained in the snowpack. SWE is the standard metric in estimating the volume of meltwater in the snowpack, which plays a critical role in the seasonal snow water supply in the RG Basin. SWE is measured at in-situ Snow Telemetry (SNOTEL) sites throughout the RG Basin and the greater Western U.S. However, the spatial distribution of SWE is complex and often difficult to estimate outside of the in-situ SNOTEL measurements. While SNOTEL provides important point-based measurements, it has been suggested that these data may not be representative of the surrounding snow-covered area (Meromy et al., 2013). Whereas regional tendencies of spatial distribution are consistent, local snowpack volumes are more variable, all of which influences the magnitude and timing of snowmelt-driven runoff.

SWE Analysis Tool

The RG Basin, and more broadly the Western U.S., is experiencing a prolonged and severe drought (EPA, 2016a), exacerbated by a warming climate that increases evapotranspiration, reduces overall snowpack quantity, and reduces the seasonal duration of snowpack. The application of the SWE Analysis Tool (SWEAT) will supply scientists and decision-makers with an improved insight into the spatial distribution of SWE by providing real-time spatial SWE estimates for the RG Basin, its subbasins, and elevation bands within the basin.

Currently, SWEAT incorporates the SNODAS gridded SWE data product (NOHRSC, 2004). Because no spaceborne sensor can directly measure SWE, spatial distribution of SWE must be estimated from remotely sensed observations coupled with modeling approaches. However, gridded SWE products, like SNODAS, are known to have accuracy issues across both mountainous and non-mountainous regions (Mudryk et al., 2025), resulting in a range of spatial SWE estimates for any given location. NM OSE and INSTAAR are working to integrate more data products to build SWEAT based on an ensemble of spatially estimated SWE. By taking an ensemble approach, the goal for SWEAT is to provide spatial SWE estimations for the RG Basin that are, on average, equally or more accurate than any individual model (Thompson, 1977; Arsenault et al., 2015). In addition, the use of a single estimate calculated from an ensemble can reduce confusion for water managers about which product may be performing best, providing a path towards consistency throughout users, and is useful in identifying product outliers.

SNODAS

The SNOw Data Assimilation System (SNODAS) represents a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center (NOHRSC). SNODAS generates spatial estimates of SWE, snow cover, and snow depth through a combination of remotely sensed snow data and in-situ measurements of SWE from SNOTEL at a 1-km spatial resolution and 24-hour temporal resolution.

The national SNODAS dataset is available through the SNODAS Data Products at National Snow and Ice Data Center (NSIDC), Version 1 (NOHRSC, 2004). SNODAS data can be viewed at the NOHRSC Interactive Snow Information page. Gridded observed snowfall images, seasonal totals, and data downloads in several formats can be found on the National Gridded Snowfall Analysis page.

Refer to the SNODAS User Guide and the NSIDC Special Report 11 (Barrett, 2003) for more information.

Discussion

Figures and tables for the identified basins (Figure 1) are presented below. The median and mean SWE (in) is presented to avoid potentially skewed outliers, most notably due to an uneven distribution of SWE across elevation bands. While the mean value represents the average value, it is highly influenced by outliers and is typically best used for uniform distributions of data. In contrast, the median value represents the middle value and is less sensitive to outliers, which can be more useful in skewed distributions. Note that the basin-wide median and mean values may not be directly comparable with the SNOTEL basin-wide values. If comparisons are made with SNOTEL sites, a better comparison may be made across elevation bands (Table 3).

Figure 3 (Rio Grande Headwaters), Figure 4 (Upper Rio Grande), and Figure 5 (Rio Grande-Elephant Butte) show the spatial distribution of SWE amount (in), change in SWE volume (af) since the last report, and SWE volume (af) across subbasin and elevation. Since the last report, spring snowmelt persisted throughout the Rio Grande Basin study area, resulting in a decrease in SWE in all three HUC6 Basins. Estimated SNODAS SWE decreased across all HUC8 subbasins and elevation bands except for the Upper Rio Grande HUC8 subbasin between 12,000 and 13,000 feet, which showed an increase of 338 af (Figure 4; Table 3). Estimated SNODAS SWE decreased to 0 af throughout the entire Rio Grande-Elephant Butte HUC6 Basin (Figure 5; Tables 1 and 3).

Estimated SWE amounts (in), volumes (af), and change in volumes (af) since the last report are presented in the tables below.

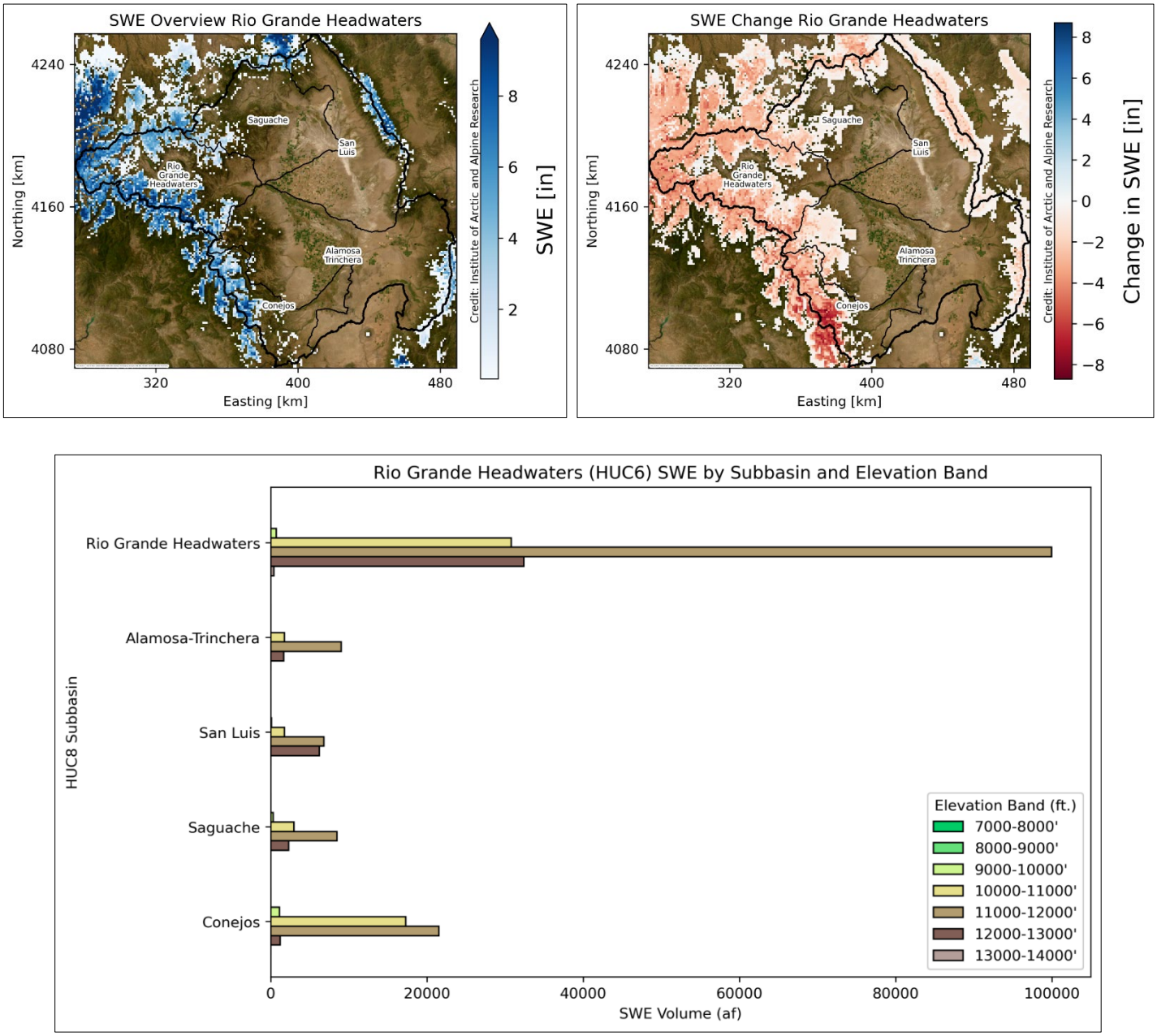

Figure 1. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande Headwaters Basin for April 29, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Rio Grande Headwaters Basin (HUC6), including the Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande Headwaters Basin (bottom).*

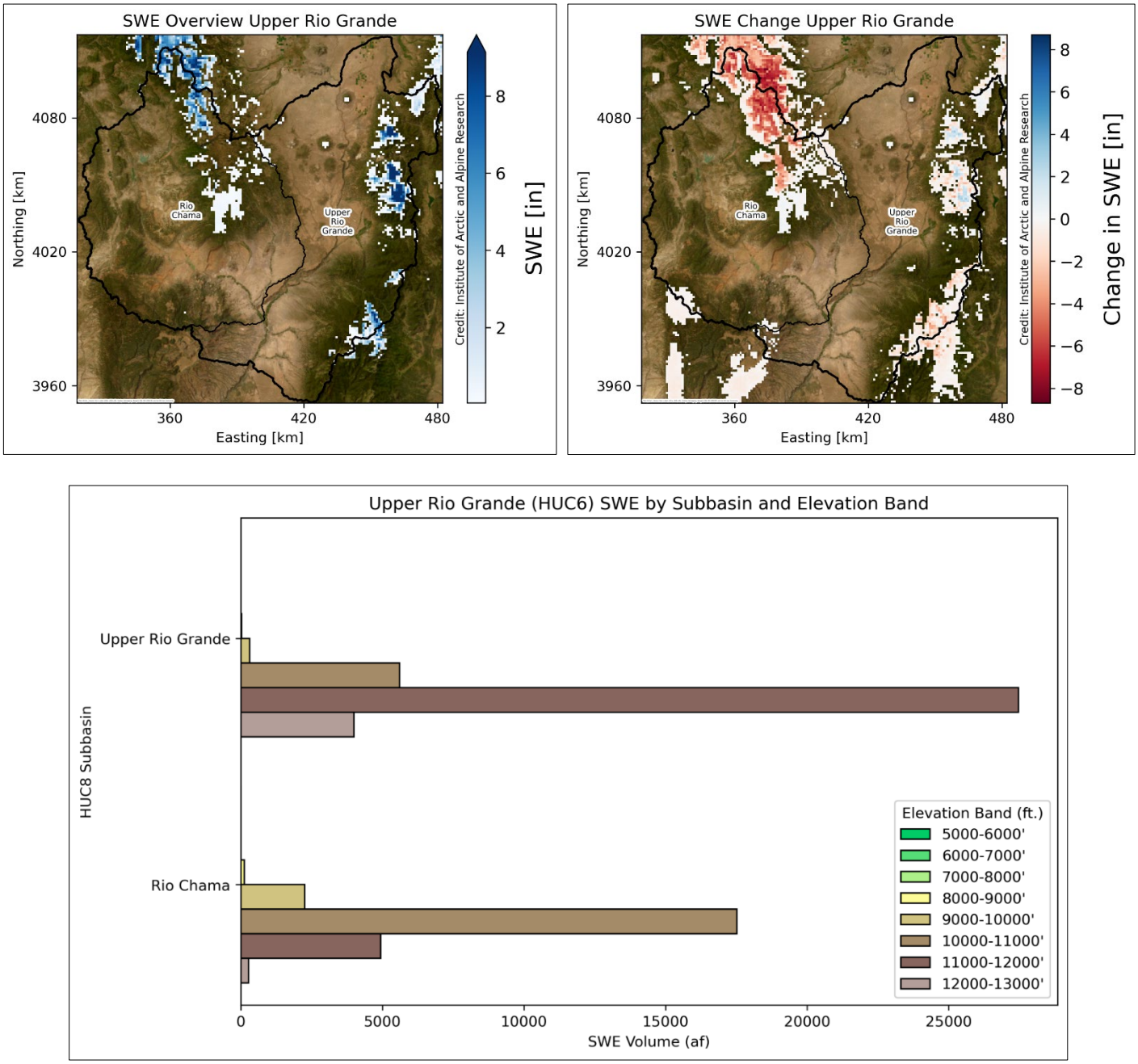

Figure 2. Estimated SNODAS SWE (in) and SWE volume (af) across the Upper Rio Grande Basin for April 29, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Upper Rio Grande Basin (HUC6), including the Upper Rio Grande and Rio Chama subbasins. SWE volume (af) by subbasin and elevation band across the Upper Rio Grande Basin (bottom).*

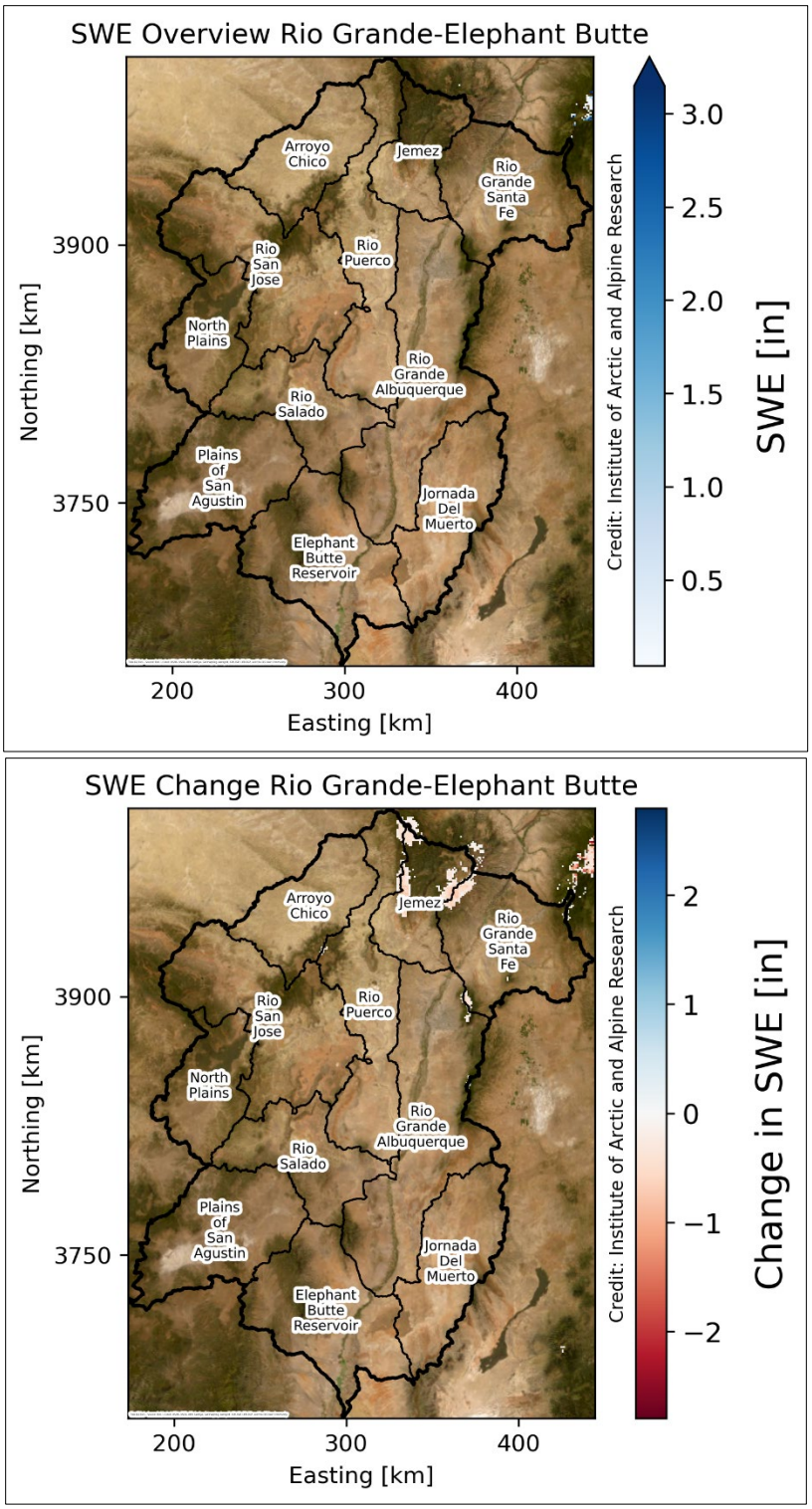

Figure 3. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande-Elephant Butte Basin for April 29, 2025. *SWE amount (top) and change in SWE since the last report (top) across Rio Grande-Elephant Butte (HUC6), including the Rio Grande-Santa Fe, Jemez, Rio Grande-Albuquerque, Arroyo Chico, North Plains, Rio San Jose, Plains of San Agustin, Rio Salado, Jornada Del Muerto, Elephant Butte Reservoir, and Rio Puerco subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande-Elephant Butte Basin is not shown due to no estimated SWE volume.*

Tables

Table 1. Estimated SWE by HUC6 Basin for April 29, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 22, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	0.0	2.2	164,386	-125,653
	Alamosa-Trinchera	0.0	0.1	12,484	-28,218
	San Luis	0.0	0.2	14,947	-16,002
	Saguache	0.0	0.2	14,101	-22,625
	Conejos	0.0	1.0	41,239	-55,359
	Total	**0.0**	**0.6**	**247,158**	**-247,857**
Upper Rio Grande	Upper Rio Grande	0.0	0.2	37,413	-9,354
	Rio Chama	0.0	0.1	25,119	-48,598
	Total	**0.0**	**0.2**	**62,532**	**-57,952**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	0.0	0.0	0	-1,503
	Jemez	0.0	0.0	0	-2,880
	Rio Grande-Albuquerque	0.0	0.0	0	-173
	Arroyo Chico	0.0	0.0	0	-18
	North Plains	0.0	0.0	0	0
	Rio San Jose	0.0	0.0	0	0
	Plains of San Agustin	0.0	0.0	0	0
	Rio Salado	0.0	0.0	0	0
	Jornada Del Muerto	0.0	0.0	0	0
	Elephant Butte Reservoir	0.0	0.0	0	0
	Rio Puerco	0.0	0.0	0	-645
	Total	**0.0**	**0.0**	**0**	**-5,219**

Table 2. Estimated SWE by Specified Subbasin Groupings for April 29, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

Subbasin Grouping	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 22, 2025 (af)
Upper Rio Grande	Rio Grande Headwaters	0.0	2.2	164,386	-125,653
	Saguache	0.0	0.2	14,101	-22,625
	Conejos	0.0	1.0	41,239	-55,359
	Total	**0.0**	**1.2**	**219,727**	**-203,637**
Sangre de Cristo	San Luis	0.0	0.2	14,947	-16,002
	Alamosa-Trinchera	0.0	0.1	12,484	-28,218
	Upper Rio Grande	0.0	0.2	37,413	-9,354
	Total	**0.0**	**0.2**	**64,844**	**-53,574**
Rio Chama		**0.0**	**0.1**	**25,119**	**-48,598**
Jemez		**0.0**	**0.0**	**0**	**-2,880**

Table 3. Estimated SWE by HUC6 Basin and Elevation Band for April 29, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by elevation band (1000 ft. increments) and HUC8 subbasin.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 22, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-84
		9000-10000'	0.0	0.1	749	-3,462
		10000-11000'	0.0	1.4	30,790	-36,762
		11000-12000'	4.8	4.6	99,973	-66,988
		12000-13000'	5.7	5.4	32,439	-18,099
		13000-14000'	4.6	3.5	435	-258
	Alamosa-Trinchera	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-118
		9000-10000'	0.0	0.0	21	-3,112
		10000-11000'	0.0	0.1	1,748	-10,842
		11000-12000'	0.2	1.1	9,034	-12,583
		12000-13000'	0.2	0.9	1,677	-1,519
		13000-14000'	0.0	0.0	3	-45
	San Luis	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-15
		9000-10000'	0.0	0.0	85	-1,115
		10000-11000'	0.0	0.3	1,775	-4,616
		11000-12000'	0.0	1.5	6,835	-6,671
		12000-13000'	1.2	2.4	6,227	-3,523
		13000-14000'	0.0	0.4	25	-62
	Saguache	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-5
		9000-10000'	0.0	0.0	330	-2,696
		10000-11000'	0.0	0.2	2,969	-8,600
		11000-12000'	1.4	1.8	8,490	-9,504
		12000-13000'	2.2	2.5	2,313	-1,808
		13000-14000'	0.0	0.0	0	-13
	Conejos	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	6	-1
		9000-10000'	0.0	0.1	1,119	-5,498
		10000-11000'	0.0	1.8	17,321	-34,542
		11000-12000'	4.6	4.4	21,550	-14,413

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 22, 2025 (af)
	Conejos	12000-13000'	3.1	3.2	1,243	-904
Upper Rio Grande	Upper Rio Grande	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	28	-38
		9000-10000'	0.0	0.0	315	-1,229
		10000-11000'	0.0	0.3	5,608	-6,894
		11000-12000'	1.0	3.5	27,467	-1,531
		12000-13000'	1.7	3.4	3,996	338
	Rio Chama	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	131	-482
		9000-10000'	0.0	0.1	2,256	-8,340
		10000-11000'	0.0	1.6	17,517	-37,312
		11000-12000'	6.2	6.1	4,936	-2,377
		12000-13000'	6.8	6.8	279	-87
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	-62
		8000-9000'	0.0	0.0	0	-1,076
		9000-10000'	0.0	0.0	0	-341
		10000-11000'	0.0	0.0	0	-24
		11000-12000'	0.0	0.0	0	0
	Jemez	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	-298
		8000-9000'	0.0	0.0	0	-1,527
		9000-10000'	0.0	0.0	0	-759
		10000-11000'	0.0	0.0	0	-295
	Rio Grande-Albuquerque	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-80

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 22, 2025 (af)
Rio Grande-Elephant Butte	Rio Grande-Albuquerque	9000-10000'	0.0	0.0	0	-73
		10000-11000'	0.0	0.0	0	-20
	Arroyo Chico	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-18
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	North Plains	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Rio San Jose	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	Plains of San Agustin	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
	Rio Salado	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
	Jornada Del Muerto	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Elephant Butte Reservoir	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 22, 2025 (af)
Rio Grande-Elephant Butte	Elephant Butte Reservoir	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0
	Rio Puerco	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-213
		9000-10000'	0.0	0.0	0	-224
		10000-11000'	0.0	0.0	0	-208

Acknowledgments

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